Corbin J. Allison

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Experience

TollWarden, LLC, tollwarden.com July 2026 - Present
Founder and Software Engineer

- Built and operate a payment security service for AI agents that pay with x402 stablecoin micropayments. Each scan returns an Ed25519-signed allow, flag, or block verdict bound to a hash of the exact payment
- Wrote detectors for prompt-injection-triggered payments, replay, address poisoning, overpayment, and secret leakage, gated in CI by an evaluation corpus of 125 attack and benign cases
- Node.js and TypeScript server with 763 automated tests, deployed on Render and published to npm and the MCP Registry, plus TypeScript and Python SDKs and integrations for LangChain, CrewAI, the Vercel AI SDK, NVIDIA NeMo, and Coinbase AgentKit
- Run five scheduled AI agents that red-team the detectors, propose detector fixes, review prepared branches, and draft documentation, each limited to its own files, with every merge approved by hand

JILA, University of Colorado Boulder, Boulder, CO October 2025 - July 2026
Graduate Researcher, Becker group

- Simulated ionization and nonlinear upconversion of argon and xenon in intense multichromatic laser fields by solving the time-dependent Schrödinger equation in Fortran 90, toward vacuum-ultraviolet excitation of the ^{229m}Th nuclear clock transition
- Built Python analysis for photoelectron spectra, angular distributions, and high-harmonic yields, and presented the results at DAMOP 2026

Kansas State University, Manhattan, KS June 2022 - August 2024
Undergraduate Researcher, Greenman group and James R. Macdonald Laboratory

- Computed attosecond-controlled ionization asymmetries in chiral molecules with Molpro and ePolyScat on NERSC Perlmutter, leading to co-authored papers in *Nature* and *Physical Review Research*
- Ran LabVIEW data acquisition for an attosecond RABBITT experiment with velocity map imaging

Heartland Electric, Inc., Fort Riley, KS May 2015 - August 2022
Electrician

- Installed, troubleshot, and repaired industrial, commercial, and residential electrical systems, including SCADA, variable frequency drive, and motor control systems

Education

University of Colorado Boulder, M.S. in Physics Expected May 2027
Coursework in quantum mechanics, statistical mechanics, quantum optics, and computational chemistry. Graduate teaching assistant since August 2024 for General Physics 1, Electronics for the Physical Sciences (lab sections and circuit debugging), and Atomic and Molecular Spectra

Kansas State University, B.S. in Physics, minors in Mathematics and Computer Science May 2024
University Honors Program thesis on attosecond control of photoionization in chiral molecules

Publications

M. Han, J.-B. Ji, A. Blech, R. E. Goetz, **C. Allison**, L. Greenman, C. P. Koch, and H. J. Wörner, Attosecond control and measurement of chiral photoionization dynamics, *Nature* **645**, 95 (2025)

R. E. Goetz, A. Blech, **C. Allison**, C. P. Koch, and L. Greenman, Continuum-electron interferometry for enhancement of photoelectron circular dichroism, *Phys. Rev. Research* **7**, L032036 (2025)

Skills

Languages	TypeScript, JavaScript, Python, Fortran, C/C++, Bash, LaTeX
AI	Claude Code and agent workflows, Model Context Protocol, LangChain, CrewAI, LLM evaluation and prompt-injection detection
Engineering	Node.js, Git, GitHub Actions release automation, Docker, Render, npm and PyPI, Ed25519 signing, HPC (NERSC)